Olivia Alvarez

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Biostatistics master's student with strong mathematics background, specializing in building data cleaning pipelines, harmonizing complex datasets, and applying advanced statistical methods. Developing proficiencies in R, Python, SQL, and Tableau. Research interests in longitudinal studies, clinical trials, and nutrition. Aiming to utilize my strong communication skills and analytical mindset to bring meaning and action to data.

EDUCATION:

Masters of Science in Biostatistics
University of Washington - Seattle, WA | September 2026 - Spring 2028 (expected)
Honors: Biostatistics Excellence Award Scholarship
Upcoming Coursework: Statistical Inference II, Computational Skills for Biostatistics (R), Statistical Computing for Biomedical Data Science (Python), Data Analysis and Reporting, Machine Learning for Biomedical and Public Health Big Data

Bachelor of Science in Mathematics| Minors: Communication, Management, Psychology | GPA: 3.83
Purdue University - West Lafayette, IN | Graduated: December 2025
Honors: Presidential Merit Scholarship Recipient

SKILLS:

Languages and Libraries: R, SAS, SQL, Python, Git, tidyverse, ggplot2
Visualization Techniques: BioRender, Tableau, Wix, Canva
Soft: Public/presentational speaking

EXPERIENCE:

Research Intern | Biomedical Engineering and Informatics Summer Research Internship
Wake Forest School of Medicine | May 2025 - August 2025

- Worked independently under faculty biostatisticians Drs. Jaime Speiser and Nate O'Connell
- Project 1:
  - Expanded capabilities of the existing open-source R package *OpenClustered* by sourcing, cleaning, validating, and integrating 18+ continuous clustered datasets from open-source repositories
  - Engineered a pipeline for visualizing and summarizing meta-data characteristics
  - Delivered a data storytelling presentation to a diverse audience, articulating how the repository bridges existing gaps in clustered data benchmarking
- Project 2:
  - Harmonized longitudinal health data in R across three independent studies with 3000+ entries to create a unified baseline dataset
  - Engineered a data cleaning pipeline to map, recode, and align 30+ conflicting baseline health variables
  - Generated output variable summaries for the integrated data to reduce future analysis preparation
- Special Projects and Leadership:
  - Facilitated and participated in discussions evaluating the ethical uses, benefits, and risks associated with Artificial Intelligence (AI) in healthcare and informatics
  - Attended Institutional Review Board (IRB) sessions to assess risks and benefits of proposed studies, ensure compliance to guidelines and regulations, and safeguard the welfare of study participants

Research Fellow | Summer Institute of Biostatistics and Data Science
Boston University School of Public Health | June 2024 - August 2024

- Selected for a competitive, fully funded program by the National Institute of Allergy and Infectious Diseases (NIAID) to receive training in epidemiological methods, clinical trial design, and data collection frameworks
- Collaborated in a small, cross-disciplinary undergraduate team under Drs. Howard Cabral and Jacqueline Hicks to guide analysis from raw data into evidence based conclusions in SAS
- Presented analytical findings on the outcomes of food insecurity from the National Health and Nutrition Examination Survey (NHANES) dataset to a panel of principal investigators and peers

PROJECTS:

Creating a Predictive Model for Housing Price in the UK

- Developed an end-to-end predictive model utilizing Multiple Linear Regression (MLR) to analyze UK housing prices
- Conducted rigorous statistical diagnostics - including residual analysis, multicollinearity checks (VIF), normality testing, outlier detection, and remedial methods - to optimize model assumptions and improve prediction accuracy
- Integrated analysis findings from four individual research questions into a cumulative final report